A FINITE TYPE MODULE FLAT ALONG A FIBRE IS FLAT ON AN OPEN AFTER AN ELEMENTARY ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The pointwise version of the flat along the fibre statement. Same chapter flat.tex, label lemma-finite-type-flat-at-point, invoked at proof line 2819. Produces, for each point of the fibre, an elementary etale neighbourhood and an open neighbourhood of the point over the base change on which the pullback of the module is flat; assembling finitely many of them is what the counted proof supplies.
- Existence of an elementary etale neighbourhood dominating finitely many given ones. More on Morphisms, label more-morphisms-lemma-elementary-etale-neighbourhoods, invoked at proof line 2832. Supplies one elementary etale neighbourhood of the base dominating the finitely many pointwise neighbourhoods, which is what makes the union of the pieces an open neighbourhood of the fibre over a single base neighbourhood.
- The finitely presented version of the same statement. Same chapter flat.tex, label lemma-finite-type-flat-along-fibre, invoked at proof line 2816. The author notes in the first sentence of the proof that the only difference from that lemma is the absence of the finite presentation hypothesis on the morphism.

Source reference. Original lemma 05KU: sources/stacks-flat/pinned-original-source.tex, source lines 2745–2745. Navigation pointer for \ref{lemma-finite-type-flat-along-fibre} used inside the counted span (line 2816); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05KT: sources/stacks-flat/pinned-original-source.tex, source lines 2701–2701. Navigation pointer for \ref{lemma-finite-type-flat-at-point} used inside the counted span (line 2819); the argument that consumes it is retained in the span.

Source reference. Original lemma 057B: sources/stacks-chow/tags.txt, source lines 6739–6739. Navigation pointer for \ref{more-morphisms-lemma-elementary-etale-neighbourhoods} used inside the counted span (line 2832); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $s \in S$. Assume that*

(1) *f is of finite type,*
(2) *$\mathcal{F}$ is of finite type, and*
(3) *$\mathcal{F}$ is flat over S at every point of the fibre X_s.*

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and an open subscheme

$$V \subset X \times_S \operatorname{Spec}(\mathcal{O}_{S',s'})$$

which contains the fibre $X_s = X \times_S s'$ such that the pullback of $\mathcal{F}$ to V is flat over $\mathcal{O}_{S',s'}$.

Proof. (The only difference between this and Lemma 05KU is that we do not assume f is of finite presentation.) For every point $x \in X_s$ we can use Lemma 05KT to find an elementary étale neighbourhood $(S_x, s_x) \to (S, s)$ and an open $V_x \subset X \times_S \operatorname{Spec}(\mathcal{O}_{S_x,s_x})$ such that $x \in X_s = X \times_S s_x$ is contained in V_x and such that the pullback of $\mathcal{F}$ to V_x is flat over $\mathcal{O}_{S_x,s_x}$. In particular we may view the fibre $(V_x)_{s_x}$ as an open neighbourhood of x in X_s. Because X_s is quasi-compact we can find a finite number of points $x_1, \ldots, x_n \in X_s$ such that X_s is the union of the $(V_{x_i})_{s_{x_i}}$. Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ which dominates each of the neighbourhoods (S_{x_i}, s_{x_i}), see More on Morphisms, Lemma 057B. Set $V = \bigcup V_i$ where V_i is the inverse images of the open V_{x_i} via the morphism

$$X \times_S \operatorname{Spec}(\mathcal{O}_{S',s'}) \longrightarrow X \times_S \operatorname{Spec}(\mathcal{O}_{S_{x_i},s_{x_i}})$$

By construction V contains X_s and by construction the pullback of $\mathcal{F}$ to V is flat over $\mathcal{O}_{S',s'}$. □